Survival-guided matrix factorization identifies reproducible prognostic programs in pancreatic cancer

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The transcriptional programs that explain the most variance in bulk tumor expression data are often not those that predict patient survival. Standard nonnegative matrix factorization (NMF) discovers latent structure by minimizing reconstruction error, which may be driven by high-variance but prognostically neutral signals such as tissue composition. Here we present DeSurv, a survival-supervised deconvolution framework that jointly integrates NMF with Cox proportional hazards modeling, shaping gene programs to capture outcome-relevant structure during discovery rather than only in post hoc evaluation. In simulations with known latent structure, DeSurv places more true prognostic genes among the top-weighted genes of each factor than standard NMF (precision, defined as the fraction of top-weighted factor genes that are true prognostic markers, 0.50 vs. 0.07), with the largest advantage when variance and prognosis diverge. In pancreatic ductal adenocarcinoma (PDAC), DeSurv recovers a compact, three-factor decomposition in which a single tumor–stroma coupling drives prognosis while exocrine-compositional signals are suppressed. These survival-aligned factors generalize to five independent PDAC cohorts without retraining, with stronger out-of-sample survival stratification than their unsupervised counterparts (pooled validation hazard ratio (HR) per standard deviation (SD) = 1.50, 95% confidence interval (CI) 1.31–1.72, P < 0.001), demonstrating that outcome supervision during factorization yields more transportable transcriptional signatures than the standard discover-then-evaluate paradigm. By prioritizing transcriptional programs that survive cross-cohort generalization, DeSurv supports the development of reproducible prognostic biomarkers and risk-stratification tools for the bulk-transcriptomic cohorts that dominate clinical cancer research.

nonnegative matrix factorization | survival analysis | semi-supervised learning | tumor deconvolution | pancreatic cancer

Identifying the transcriptional programs that predict clinical outcomes is central to molecular subtyping and prognostic stratification in cancer (1). Bulk transcriptomic profiling remains the dominant technology for the large, clinically annotated cohorts required for such survival modeling (2, 3). Single-cell and spatial atlases clarify the cellular states present in PDAC, but cohort-scale survival annotation remains largely bulk-transcriptomic; supervised decomposition of bulk mixtures therefore remains necessary for population-scale prognostic discovery. Each expression profile is a composite of malignant cells, cancer-associated fibroblasts (CAFs), immune infiltrates, and other microenvironmental components (4). As a result, these signatures may be diluted or obscured by variation in cellular composition across samples. Nonnegative matrix factorization (NMF) has been instrumental in resolving this mixture into additive, interpretable gene programs corresponding to recognizable cell types or transcriptional states (5–7). In pancreatic ductal adenocarcinoma (PDAC), this approach has yielded key biological insights, including the identification of molecular subtypes (1, 8–10), the Basal-like and Classical tumor programs defined through virtual microdissection (6), and compartment-specific deconvolution of tumor and microenvironmental signals (7).

However, the standard approach discovers these programs through unsupervised factorization (11) and only then evaluates their clinical relevance retrospectively (1, 8). Because unsupervised NMF minimizes reconstruction error, its factors tend to be dominated by the highest-variance patterns in the data, which may reflect tissue composition or other outcome-neutral sources rather than prognostic biology (2, 12). In PDAC, tissue-composition signals such as from exocrine cells reflect low tumor purity rather than cancer cell-intrinsic programs and carry little prognostic value (9, 10). This misalignment between variance and prognosis extends beyond PDAC: tumor purity alone accounts for a large fraction of expression variation across cancer types (13), yet the objective optimized during unsupervised discovery (reconstruction error)

Significance Statement

Bulk tumor samples mix cancer cells with surrounding normal cells, obscuring the genes that truly predict patient outcomes. Existing deconvolution methods discover latent gene programs but do not ensure prognostic relevance, while supervised predictors often sacrifice biological interpretability. We present DeSurv, a survival-supervised deconvolution framework that integrates nonnegative matrix factorization with Cox modeling to learn gene programs and their survival associations jointly. By embedding outcome information into discovery, DeSurv yields clinically relevant, reproducible programs across cohorts. This provides a general framework for identifying prognostically informative gene programs without requiring post hoc filtering or pre-specified signatures. Such reproducible programs are essential for risk stratification, clinical-trial enrichment, and therapeutic-target prioritization in the bulk-transcriptomic cohorts that dominate clinical cancer research.

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differs fundamentally from the criterion used during evaluation (survival association) (12). Identifying the prognostically relevant programs among the discovered factors requires extensive downstream evaluation: over a decade of PDAC subtyping was needed to converge on a robust Basal-like/Classical dichotomy, first defined through virtual microdissection (6) and subsequently confirmed in independent cohorts (9), single-cell analyses (14, 15), and prospective clinical profiling (3).

Supervised dimension reduction theory establishes that response-guided subspace estimation targets directions most relevant to the outcome, whereas variance-maximizing projections can miss outcome-relevant structure entirely (12, 16), and outcome-weighted methods have demonstrated this advantage in clustering and principal component settings (2, 17). Whether this advantage extends to NMF-based deconvolution, where nonnegative constraints restrict the factor space geometry, censored survival outcomes provide a noisier supervisory signal, and overly aggressive survival gradients could compromise the mixture-coefficient interpretation of sample loadings, remains an open empirical question. The present work provides empirical evidence that survival-guided NMF recovers prognostic programs more reliably than unsupervised factorization in this setting. This advantage is established through two non-circular tests: recovery of ground-truth program identities in simulations where the true gene sets are known a priori, and out-of-sample generalization to five independent cohorts on which no model parameters were estimated.

Here we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox proportional hazards modeling. DeSurv approximates the expression matrix as $X \approx WH$, where columns of W define nonnegative gene programs and rows of H encode sample-level loadings. The key architectural choice is that the survival gradient acts on the gene program matrix W but not on the sample loadings H, directing gene programs toward outcome-relevant structure while preserving the interpretation of loadings as mixture coefficients (5, 18). Because W defines shared transcriptomic programs, new samples can be scored by projection ($Z_{\text{new}} = \widetilde{W}^{\top} X_{\text{new}}$) without requiring re-optimization, a property not shared by methods that route supervision through H (19, 20). Model complexity and supervision strength are jointly selected via cross-validated concordance (Methods).

We evaluate DeSurv in two settings. In simulations with known latent structure and survival associations, DeSurv recovers the true factorization rank and prognostic programs more reliably than standard NMF. Critically, its advantage scales with the gap between the highest-variance and most prognostic subspaces and vanishes under null conditions where no survival signal exists, confirming that the method adapts to the structure of the data rather than imposing spurious survival associations. In PDAC, survival supervision produces a different factor structure: DeSurv concentrates survival signal into a single factor coupling Classical tumor identity with inflammatory CAF (iCAF)-associated stroma, a prognostic axis that unsupervised methods distribute across multiple factors, while the exocrine-compositional variation that dominates standard NMF is not represented as a dedicated factor. These survival-aligned factors generalize to five independent PDAC cohorts without retraining, with stronger out-of-sample survival stratification than their unsupervised counterparts. Because the misalignment between reconstruction-optimal and outcome-relevant subspaces is not unique to PDAC (tumor purity variation dominates expression landscapes across cancer types), DeSurv provides a general framework for extracting outcome-relevant programs from bulk tumor transcriptomes wherever unsupervised factors conflate biological signal with compositional noise. Survival-supervised cross-validation simultaneously resolves the rank-selection ambiguity that unsupervised diagnostics leave open, offering a route to reproducible prognostic biomarkers and biological insight into tumor–stroma coupling in cancers like PDAC where compositional variation dominates expression. The method is implemented in an open-source R package (https://github.com/rashidlab/DeSurv).

Results

Model Overview. We developed DeSurv, a survival-supervised deconvolution framework (Fig. 1; Methods), and evaluated it in controlled simulations and PDAC cohort analysis with external validation.

Survival supervision clarifies NMF rank selection. We first asked whether incorporating survival outcomes resolves the instability of unsupervised rank selection heuristics (11, 21). Using gene expression data from the PDAC training cohorts (The Cancer Genome Atlas (TCGA) (22) and the Clinical Proteomic Tumor Analysis Consortium (CPTAC) (23); Materials and Methods), which consist of non-metastatic, treatment-naive specimens (SI Appendix, Table S1), we evaluated commonly used unsupervised criteria across a range of candidate ranks.

Standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks — a known limitation of unsupervised rank-selection criteria, which can disagree on the same dataset (11, 21) (SI Appendix, Fig. S4).

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance: cross-validated concordance index (C-index), optimized jointly with supervision strength α via Bayesian Optimization (BO). The resulting C-index surface identified a coherent region of high performance — a broad plateau spanning $k = 3$ through $k = 12$ along DeSurv's supervised dimension (Fig. 2D), indicating that three supervised factors capture nearly as much prognostic information as twelve. Model selection followed the one-standard-error rule (Methods): we selected the smallest k whose predicted performance lay within one standard error of the maximum, yielding a parsimonious choice. Applying the 1-SE rule to the C-index surface selected $k = 3$ and $\alpha = 0.334$ (peak C-index 0.655; $n = 273$ patients, 139 events). Survival-supervised model selection thus resolves the rank ambiguity that unsupervised heuristics leave open, yielding a single parsimonious factorization for downstream evaluation.

DeSurv recovers prognostic gene programs when variance and prognosis diverge. To test whether survival supervision improves gene program recovery under nonnegative constraints, we designed simulations with known ground-truth latent structure. Simulated expression matrices were generated from a nonnegative factor model in which prognostic gene programs explained low variance relative to outcome-neutral background signals, with survival times generated from latent factor scores via a Cox model (p = 3,000 genes, n = 200 samples, true k = 3; Materials and Methods, Simulation Studies). To test whether survival information should enter during factorization rather than only afterward, we compared DeSurv to standard NMF ($\alpha = 0$), where factors are learned by reconstruction error alone and survival associations are estimated by a separately fit Cox model. Both methods were tuned via BO over cross-validated C-index, so that differences in performance reflect the role of supervision during factorization, not during model selection.

In the primary scenario, where prognostic programs explained low variance relative to outcome-neutral programs, DeSurv consistently achieved higher C-index (Fig. 2A) and substantially higher precision for recovering the true prognostic gene programs (Fig. 2B). Here precision denotes the fraction of a factor's top-weighted genes — those with the largest factor specificity score s_{ij} (SI Appendix, Section 9), with n_{top} chosen by Bayesian optimization (SI Appendix, Section 6) — that overlap with a true prognostic program's gene set. The unsupervised baseline exhibited near-zero precision, indicating that variance-driven factorization failed to concentrate on the genes that actually drove survival. This confirms that outcome supervision during factorization is necessary to recover prognostic programs that do not dominate expression variance. Across 100 replicates, DeSurv achieved median C-index 0.839 versus 0.724 for standard NMF ($\Delta = 0.115$, paired Wilcoxon $P < 0.001$). The smaller C-index gap reflects what each metric measures: NMF factor scores aggregate many genes, so partial marker recovery at moderate loadings preserves moderate within-cohort C-index even when top-weighted positions are dominated by outcome-neutral background — a gene-attribution failure that precision exposes. Cross-cohort transportability depends on the biological stability of the factors themselves and is assessed in the PDAC external validation analyses.

To verify that these gains reflect genuine signal recovery rather than overfitting, we repeated the analysis under a null scenario ($\beta = 0$) and a mixed scenario with partial overlap between variance-dominant and prognostic structure (SI Appendix, Fig. S2). DeSurv's benefit scaled with the degree to which prognostic programs were masked by outcome-neutral variation: largest in the primary scenario ($\Delta = 0.115$), attenuated when variance and prognosis partially overlapped ($\Delta = 0.065$, paired Wilcoxon $P < 0.001$), and absent when no survival signal existed, where both methods yielded C-index near 0.5. DeSurv also recovered the true rank ($k = 3$) more frequently than standard NMF, which systematically under-selected near $k = 2$ (Fig. 2C).

Survival supervision and standard NMF produce different factor structures in PDAC. BO selected $k = 3$ and $\alpha = 0.334$ for DeSurv in the PDAC training cohorts (TCGA and CPTAC). To compare the factor structures produced with and without survival supervision at the same dimensionality, we fit standard NMF at the same rank ($k = 3$) and examined the overlap between each method's factor-specific gene rankings and established PDAC gene programs (Fig. 3A–B). Matching k across methods isolates the effect of survival supervision on factor content from differences in model complexity; standard NMF's own rank-selection diagnostics did not converge on a clear alternative (SI Appendix, Fig. S4), and the sensitivity analysis below (and SI Appendix, Table S5) evaluates NMF performance across a range of ranks and supervision strengths. We first describe the biological identity of each method's factors, then ask whether these structural differences correspond to differences in survival contribution.

We refer to the three DeSurv factors as D1–D3 and the three NMF factors as N1–N3. In the DeSurv factorization, D1 correlated with both Classical tumor programs and iCAF-associated stromal signatures, D2 with myofibroblastic CAF (myCAF)-associated (activated immune-stromal) programs, and D3 with Basal-like tumor programs (Fig. 3A). The co-occurrence of Classical and iCAF-associated signatures within a single factor is consistent with evidence that survival stratification in PDAC improves when tumor-intrinsic and CAF subtypes are considered jointly (10, 24). No DeSurv factor showed a significant positive correlation with exocrine-associated gene programs (DECODER Exocrine, Collisson Exocrine, Bailey aberrantly differentiated endocrine–exocrine (ADEX); Fig. 3A). The survival contributions of these factors and their relationship to expression variance are quantified below.

Standard NMF produced a different factor organization (Fig. 3B). One factor (N2) was strongly correlated with exocrine-associated gene programs and negatively correlated with immune and stromal signatures (Fig. 3B). A second factor (N1) correlated with Classical tumor identity (Fig. 3B). A third factor (N3) correlated with immune, fibroblast, and extracellular matrix (ECM)-related programs (including Puleo Desmoplastic, SCISSORS pan-CAF (panCAF) and myofibroblastic CAF (myCAF)-like (25), Maurer Immune and ECM (26), and DECODER Activated Stroma), aggregating these into a single factor rather than separating them as DeSurv does (D1 and D2; Fig. 3A). At $k = 3$, NMF thus devotes one of its three factors to exocrine-associated variation, leaving two factors to capture all tumor-intrinsic and microenvironmental programs. Whether these structural differences correspond to differences in prognostic information is quantified next.

We quantified per-factor contribution using two metrics (Fig. 3C; SI Appendix, Sections 10–11). For each factor, we computed the conditional variance explained (semi-partial R^2: the incremental fraction of total variance accounted for by that factor given all other factors), and its survival contribution, measured as the change in Cox partial log-likelihood ($\Delta\ell$) when that factor is dropped from a k-factor Cox model. All three NMF factors had negligible $\Delta\ell$ despite spanning a range of variance, while DeSurv concentrated nearly all survival contribution into D1. These per-factor contrasts reflect overall model-level differences: at matched rank, the NMF $k = 3$ Cox model achieved a cross-validated C-index of 0.55 versus 0.65 for DeSurv, confirming that NMF's near-zero per-factor $\Delta\ell$ reflects weak model-level prognostic content rather than redistributed contribution across correlated factors. Specifically, D1 explained 7% of variance with $\Delta\ell = 66.4$, while the highest-variance NMF factor explained 24.3% of expression variance but contributed only $\Delta\ell = 0.1$ to survival. The highest-variance DeSurv factor (D3, 35.4% of variance) also contributed minimally to survival after accounting for D1. Thus, in PDAC the highest-variance factors are not the most prognostic, consistent with tumor purity analyses across cancer types (13).

To further characterize how the two factorizations relate, we examined pairwise W-matrix gene loading correlations (Fig. 3D). Two clear correspondences emerged: NMF's tumor-identity factor (N1, top-correlated with Classical programs) mapped strongly to DeSurv's tumor factor (D3, top-correlated with Basal-like programs), reflecting shared pan-tumor gene loadings rather than subtype concordance, and NMF's microenvironmental factor (N3) mapped primarily to DeSurv's D2. Notably, DeSurv's most prognostic factor (D1) showed only weak, roughly uniform correlations with all three NMF factors, indicating that the gene program structure underlying D1 does not have a direct counterpart in the unsupervised factorization. Because D1 couples Classical tumor and iCAF-associated stromal signatures into a single survival-aligned program, its signal is distributed across multiple NMF factors, none of which individually captures the same tumor–stroma coupling. NMF's exocrine-associated factor (N2) similarly lacked a strong DeSurv counterpart, with its strongest (though weak) correlation to D3, consistent with D3's high variance and low survival contribution. Together, the two methods recover overlapping tumor and microenvironmental structure but differ in two key respects: NMF dedicates a factor to exocrine-compositional variation that DeSurv suppresses, while DeSurv constructs a prognostic tumor–stroma factor that NMF does not isolate. The net result is that DeSurv concentrates survival contribution into a single factor ($\Delta\ell = 66.4$), whereas no NMF factor achieves comparable survival contribution at the same rank. Whether this translates to improved generalization across independent cohorts is tested below.

Survival-aligned programs generalize across independent PDAC cohorts. A prognostic factorization is useful only if it generalizes beyond the training cohort. Having established that survival supervision and standard NMF produce different factor structures in the training data, we next tested whether each method's structure transfers to independent cohorts without retraining (DeSurv at $k = 3$; NMF at $k = 3$, 5, and 7). The five external validation cohorts (Dijk, Moffitt, PACA-AU array and RNA-seq, and Puleo) likewise consist of non-metastatic, treatment-naive specimens (SI Appendix, Table S1). We computed factor scores in each validation dataset by projecting the gene programs learned in the training cohort ($Z = \widetilde{W}^\top X_{\text{new}}$), yielding a continuous score for each factor per sample.

External validation C-index for NMF at independently selected ranks is summarized in SI Appendix, Table S5. Standard NMF at elbow-selected $k = 5$ and BO-selected $k = 7$ substantially improved over NMF at $k = 3$, with $k = 7$ matching or exceeding DeSurv's per-cohort C-index in most cohorts — though only when DeSurv's tuning framework is applied ($\alpha = 0$ in the joint BO over rank and hyperparameters), since NMF's own rank-selection heuristics (cophenetic, silhouette, residual; SI Appendix, Fig. S4) do not converge on this rank. However, this concordance gain still requires more than twice as many factors. DeSurv at $k = 3$ retains a significant hazard ratio after adjustment for PurIST and DeCAF (SI Appendix, Table S6), achieving independent prognostic value more parsimoniously. Although DeSurv was trained without molecular subtype labels, the resulting high-risk group was enriched for the two established poor-prognosis subtypes (Basal-like by PurIST, proCAF by DeCAF; Fisher's exact, SI Appendix, Fig. S7) — capturing established consensus biology while adding independent prognostic information. Across 5 independent PDAC cohorts ($n = 616$), the DeSurv linear predictor showed a consistent survival association (stratified Cox model: pooled HR per SD 1.50; 95% CI 1.31–1.72; $P = 2.6 \times 10^{-9}$; per-cohort forest plot in Fig. 4A; KM curves in SI Appendix, Fig. S6C–F). At the matched rank ($k = 3$), the analogous NMF linear predictor showed a weaker pooled effect (HR per SD 1.11; 95% CI 1.00–1.23; $P = 0.057$); a direct comparison using dichotomized risk groups is presented below.

To assess clinical interpretability of these continuous associations, we dichotomized validation samples into high- and low-risk groups using method-specific cutpoints, each independently optimized via cross-validation on the training data. The DeSurv linear predictor separated survival trajectories in the pooled validation cohort (HR = 2.01, 95% CI 1.63–2.47, $P < 0.001$; Fig. 4B). The same procedure applied to NMF at $k = 3$ yielded weaker stratification (HR = 1.48, 95% CI 1.04–2.11, $P = 0.028$; Fig. 4C). The DeSurv cutpoint identified a substantially larger high-risk group than NMF (184 vs 46 high-risk patients; 138 reclassified), thereby providing finer prognostic discrimination. Together, these results indicate that incorporating survival information during factorization yields gene programs with more transportable prognostic associations than the standard discover-then-evaluate approach.

Factorization rank k=3 is robust across supervision settings. The 1-SE model selection rule chose $k = 3$ over the global BO maximum ($k = 7$; CV C-index 0.655 versus 0.646; margin 0.009). To assess whether this reflects a robust solution or an artifact of a specific tuning configuration, we conducted a sensitivity analysis across $k \in \{2, 3, 5, 7, 9\}$ and $\alpha \in \{0, 0.25, 0.35, 0.55, 0.75, 0.85, 0.95\}$ (35 combinations), holding all other hyperparameters at their BO-selected values and evaluating each combination in independent external validation cohorts via stratified Cox regression (SI Appendix, Tables S2–S4). $k = 3$ achieved significance ($P < 0.05$) across 6 of 7 supervision strengths, and $k = 7$ at a comparable 5 of 7. After additional adjustment for PurIST and DeCAF molecular classifiers, $k = 3$ retained significance at 5 of 7 supervision strengths, while $k = 7$ dropped to 2 of 7 (SI Appendix, Table S3). The parsimonious $k = 3$ solution thus validates robustly across supervision strengths and retains prognostic value beyond known classifiers, providing empirical support for the 1-SE rule. Standard NMF at $k = 5$ also achieves external significance (PurIST/DeCAF-adjusted $P = 0.0005$), providing convergent evidence for the iCAF biology that DeSurv concentrates at $k = 3$ (SI Appendix, Section 16). The supervised $k = 3$ solution achieves the same biology more parsimoniously: the additional factors at $k = 5$ capture transcriptional variance without new factor-level prognostic contribution (SI Appendix, Fig. S10). DeSurv's contribution is therefore not unconditional dominance over every NMF rank, but a supervised selection framework that identifies a compact, interpretable, externally validated solution without relying on post hoc inspection of multiple unsupervised ranks.

Discussion

Incorporating survival outcomes directly into NMF factorization reorganizes the transcriptional landscape of PDAC in a clinically meaningful way. DeSurv concentrates prognostic signal into fewer, more interpretable gene programs that generalize across independent cohorts without retraining. More broadly, the principle that outcome-guided dimensionality reduction targets different subspaces than variance-driven reduction extends beyond cancer genomics: sufficient dimension reduction theory (12) and the information bottleneck framework (27) both formalize this intuition. The empirical question framed in the Introduction — whether this principle survives the constraints of nonnegativity, censored survival outcomes, and mixture-coefficient interpretability that bulk-tumor deconvolution imposes — is resolved here in the affirmative. DeSurv's architecture (supervision on W, mixture interpretation preserved on H) enables single-sample scoring by projection, a transportability property not shared by methods that route supervision through H (19, 20).

In PDAC, the factor structure produced by survival supervision is biologically informative. The factor with the largest survival contribution (D1) couples Classical tumor programs with a iCAF-associated stromal signature, consistent with growing evidence that clinical outcomes depend on the joint configuration of tumor-intrinsic and cancer-associated fibroblast subtypes (10, 24). This co-occurrence within a single factor suggests that the prognostically relevant biology in PDAC is not purely tumor-intrinsic but reflects tumor–stroma coupling that unsupervised methods distribute across multiple factors. Standard NMF, by contrast, allocated a dedicated factor to exocrine-compositional variation, a signal that reflects tumor purity rather than cancer biology (9), leaving less capacity for the microenvironmental programs that DeSurv separates at the matched rank ($k = 3$). The methodological contribution lies not in the identity of these programs, which have been established through virtual microdissection (6), experimental microdissection (26), and unsupervised deconvolution (7), but in how they are recovered: de novo, without pre-specified signatures, with direct quantification of their survival associations during factorization.

Survival supervision concentrates prognostic signal into fewer factors, creating a broad concordance plateau that the one-standard-error rule leverages to select parsimonious ranks — a deliberate balance between biological completeness and clinical relevance. In PDAC, this plateau spanned $k = 3$–12 (Fig. 2D); for standard NMF, concordance increased steadily with k, approaching DeSurv's level by $k = 7$ (SI Appendix, Table S5) but requiring more than twice as many factors and producing a fragmented structure. Notably, NMF's own unsupervised rank selection heuristics — reconstruction residuals, cophenetic correlation, and silhouette width (SI Appendix, Fig. S4) — do not point to $k = 7$; the elbow criterion suggests $k = 5$, which does not match DeSurv's concordance. NMF only reaches $k = 7$ through Bayesian optimization with $\alpha = 0$, that is, by applying DeSurv's tuning infrastructure to an unsupervised model. Beyond this parsimony, DeSurv at $k = 3$ achieves independent prognostic value: both methods retain significance after classifier adjustment (SI Appendix, Table S6), and $k = 3$ is more robust across supervision strengths (5 of 7 versus 2 of 7 for $k = 7$; SI Appendix, Table S3). The model selection framework — joint BO over rank and hyperparameters with the 1-SE rule — is itself part of the methodological contribution, not merely the survival gradient: parsimony is a property of the 1-SE rule, but supervision creates the flat concordance surface under which the rule can operate effectively.

Our simulations provide further guidance on when this tradeoff is most beneficial: DeSurv's advantage is greatest when prognostic programs explain modest variance relative to outcome-neutral signals and vanishes under null conditions where no survival signal exists. DeSurv is therefore not intended to replace unsupervised NMF in all settings. When the goal is exploratory discovery without clinical endpoints, when survival data is sparse or unreliable, or when the application requires comprehensive biological characterization rather than prognostic stratification, unsupervised methods remain appropriate. The α parameter, tuned via Bayesian optimization, lets the data determine the appropriate balance for a given cancer type and cohort size, and the $\alpha = 0$ endpoint recovers standard NMF as a special case. DeSurv is likewise not proposed as a competing molecular classifier; translating gene programs into discrete clinical subtype labels would be a downstream application. The relevant comparison with existing classifiers is whether DeSurv's factors capture independent prognostic signal beyond what those classifiers already explain — Table S6 confirms this directly, with the DeSurv $k = 3$ linear predictor retaining a significant adjusted hazard ratio after conditioning on PurIST and DeCAF (HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$).

Several limitations define the scope of these findings. First, DeSurv assumes Cox proportional hazards, which may not hold in settings with delayed or non-proportional treatment effects; the convergence guarantee (SI Appendix, Theorem 1) applies to an idealized algorithm, and the implementation includes practical modifications that depart from the theoretical analysis, though empirically these do not affect convergence behavior. Extensions to more flexible survival models are a natural direction. Second, DeSurv was applied here only to PDAC, and broader benchmarking across cancer types is needed to establish the generality of the gap between reconstruction-optimal and survival-optimal subspaces; the computational cost of Bayesian optimization may limit scalability to very large cohorts, though hyperparameter selection is performed once and the final model applies by projection. Third, all cohorts analyzed here consist of non-metastatic, treatment-naive PDAC specimens. This homogeneity reduces confounding from treatment effects during factorization but limits direct applicability to treated patient populations where neoadjuvant chemotherapy is increasingly standard practice. Whether the gene programs identified here retain prognostic value in treated cohorts — or whether survival supervision in that setting recovers distinct treatment-responsive programs — remains an open question. Extending DeSurv to cohorts with neoadjuvant or multimodal therapy is therefore a natural and clinically important next step. Bulk profiling continues to underpin clinical molecular stratification — from trial programs such as COMPASS (3) to CLIA-validated diagnostic assays in routine clinical care — a deployment landscape favored by the cost, FFPE compatibility, and standardization requirements of the clinical workflow. As single-cell cohorts with clinical annotation grow, DeSurv's gene programs, which align with cell-type signatures confirmed by independent single-cell studies (14, 15), can serve as a bridge between bulk-derived prognostic models and emerging cellular-resolution datasets.

The standard workflow in cancer transcriptomics — unsupervised discovery followed by retrospective evaluation — incurs a structural cost whenever the highest-variance signals are not the most prognostic, likely the norm in cancer types where tumor purity and tissue composition dominate expression variation (13). DeSurv resolves this by aligning factorization with the clinical question from the outset: in PDAC, this recovered a Classical-tumor + iCAF-coupling program that transferred to five independent cohorts (pooled HR per SD 1.50) at a parsimonious $k = 3$ that survives PurIST/DeCAF adjustment. The combination of a survival-supervised gradient and joint Bayesian optimization with the one-standard-error rule provides a principled framework for extracting outcome-relevant programs from bulk tumor transcriptomes wherever clinical outcomes are available — a setting that, as clinically annotated cohorts accumulate, is more likely to expand alongside than be displaced by single-cell technologies.

Materials and Methods

A. Problem formulation and notation. Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p\times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k\times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the censoring indicators ($\delta_i = 1$ if the event is observed), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k\times n}$ and linear predictor $Z^\top\beta$.

B. The DeSurv Model. DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with patient survival. The method operates in a semi-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1-\alpha)\,\mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\,\mathcal{L}_{\mathrm{Cox}}(W, \beta), \qquad [1]$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. A critical architectural choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients (5, 18). When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF. The Cox component also accommodates additional sample-level covariates (e.g., tumor stage or grade) alongside Z, enabling adjustment for known prognostic factors during optimization.

Optimization alternates updates for H, W, and β (Algorithm 1). The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Although the joint problem is non-convex, the alternating updates converge to a stationary point under mild regularity conditions (SI Appendix, Theorem 1). Closed-form update equations, the convergence proof, and BO control parameters are given in SI Appendix, Sections 1–3.

Algorithm 1 DeSurv block-coordinate descent

Input: Expression matrix $X \in \mathbb{R}_{\geq 0}^{p\times n}$, survival (y, δ), supervision strength $\alpha \in [0, 1)$, rank k, regularization (λ, ξ), tolerance ϵ
Output: $W \in \mathbb{R}_{\geq 0}^{p\times k}$, $H \in \mathbb{R}_{\geq 0}^{k\times n}$, $\beta \in \mathbb{R}^k$
Initialize $W^{(0)}$ via consensus-based seeding (SI Appendix, Section 6); set $H^{(0)}, \beta^{(0)}$
repeat
  $H \leftarrow$ multiplicative-update minimizer of $\mathcal{L}_{\mathrm{NMF}}(W, \cdot)$
  $W \leftarrow$ multiplicative update with combined gradient $\nabla\mathcal{L} = (1-\alpha)\,\nabla_W\mathcal{L}_{\mathrm{NMF}} - \alpha\,\nabla_W\mathcal{L}_{\mathrm{Cox}}(W, \beta)$
  $\beta \leftarrow$ Coxnet coordinate descent on $Z = W^\top X$ with penalty (λ, ξ)
until $|\mathcal{L}^{(t)} - \mathcal{L}^{(t-1)}|/|\mathcal{L}^{(t-1)}| < \epsilon$

C. Hyperparameter selection and cross-validation. Hyperparameters $(k, \alpha, \lambda, \xi, n_{\mathrm{top}})$ were selected by maximizing the cross-validated C-index using Bayesian optimization (BO) (28, 29), which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lay within one standard error of the maximum). All model selection was performed entirely within training data using nested cross-validation; no validation cohort data were used during any stage of tuning. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (SI Appendix, Section 6). Before validation, each column of W was truncated to its BO-selected number of top genes (details in SI Appendix, Section 6), denoted $\widetilde{W}$. In PDAC, BO selected $n_{\mathrm{top}} = 270$ top genes per factor. A $k \times \alpha$ sensitivity analysis confirmed the robustness of the selected rank across supervision strengths (see Results and SI Appendix, Tables S2–S4).

External validation was performed by projecting new datasets onto the learned programs via $Z = \widetilde{W}^\top X_{\mathrm{new}}$ and evaluating survival associations using C-index and log-rank statistics. Because the learned W is fixed at training time and validation samples are scored by simple projection, no retraining or access to validation survival data is required. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in SI Appendix, Part I.

D. Simulation studies. Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\mathrm{Gamma}(\mathrm{shape} = 3, \mathrm{rate} = 0.8)$ and cross-loadings onto other factors from $\mathrm{Gamma}(1, 20)$; background gene loadings from

Gamma(2, 1) across all factors; and noise gene loadings from Gamma(1, 20). Sample-level factor activities H were drawn from Gamma(2, 2). Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the primary scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T\widetilde{W}$, where $\widetilde{W}$ retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a primary scenario where prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

E. Real-world datasets. We analyzed publicly available RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and five were used for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo). To harmonize differences in scale across cohorts, filtered gene expression data were within-subject rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility). More details about the datasets can be found in SI Appendix, Section 7.

Per-cohort participant flow is summarized in SI Appendix, Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, or quality-control failure, yielding 273 analytic training samples (139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation samples (414 events) across five cohorts: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181). With 139 events and 3 factor-score covariates (one per DeSurv factor at $k = 3$), the Cox-stage events-per-variable ratio is 46, well above the conventional 10-EPV threshold, supporting estimation stability.

F. Data, Materials, and Software Availability. All datasets used in this study are publicly available and de-identified; no Institutional Review Board (IRB) approval was required. Training data were obtained from the Genomic Data Commons: TCGA-PAAD (30) and CPTAC-3 (31). Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830 (32); Puleo, E-MTAB-6134 (33)), the Gene Expression Omnibus (GEO; Moffitt, GSE71729 (34)), and the International Cancer Genome Consortium (ICGC) Data Portal (PACA-AU array and RNA-seq; European Genome–phenome Archive (EGA) study EGAS00001000154 (35)). Cohort details, sample sizes after quality control, and patient-level metadata used are summarized in SI Appendix, Section 7. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study (10).

All analyses were performed in R (version 4.6.0; R Core Team) using the following key packages: DeSurv (v1.0.1; this work), `NMF` (18), `glmnet` (36), `survival`, `ggplot2`, and `cowplot`. Bayesian optimization for hyperparameter selection was implemented within DeSurv. An R package implementing DeSurv is available at github.com/rashidlab/DeSurv (archived at Zenodo: DOI `[TO BE MINTED]`). Code, processed data, and the exact session-info file (R + package versions used to render this manuscript) are available at github.com/rashidlab/DeSurv-paper (archived at Zenodo: DOI `[TO BE MINTED]`).

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Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies, deposited to GEO and ArrayExpress) and our access was limited to the de-identified expression and survival data they provide; we did not stratify analyses by sex, race, or ethnicity because consistent demographic annotations were not available across all cohorts. Sex and other demographic variables that could modify the prognostic relevance of the gene programs identified here remain important directions for follow-up work as more comprehensively annotated cohorts become available. The author team includes investigators at multiple career stages, with leadership and analytic contributions from both early-career and senior researchers across statistical, computational, and clinical disciplines.

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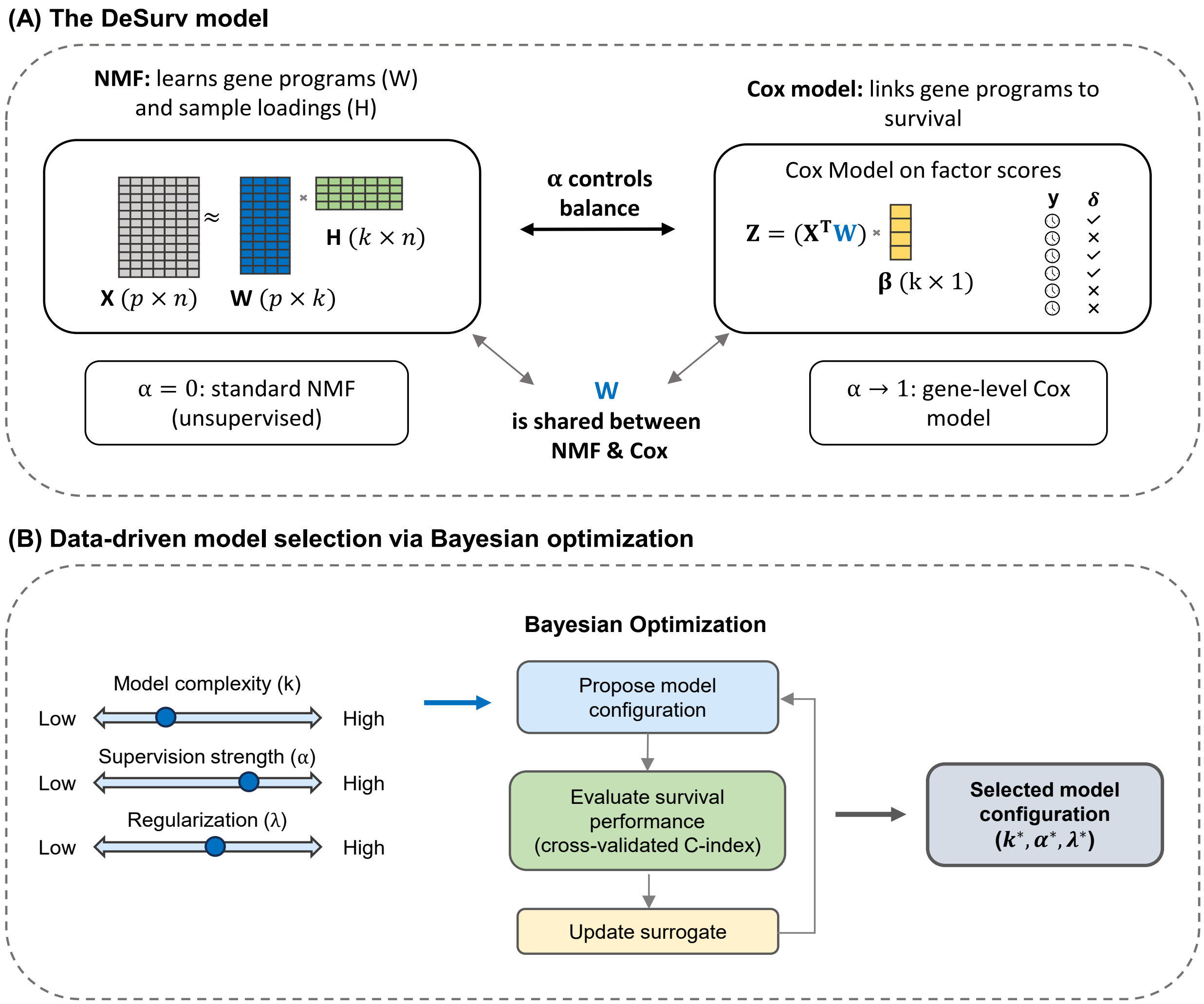

Fig. 1. Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

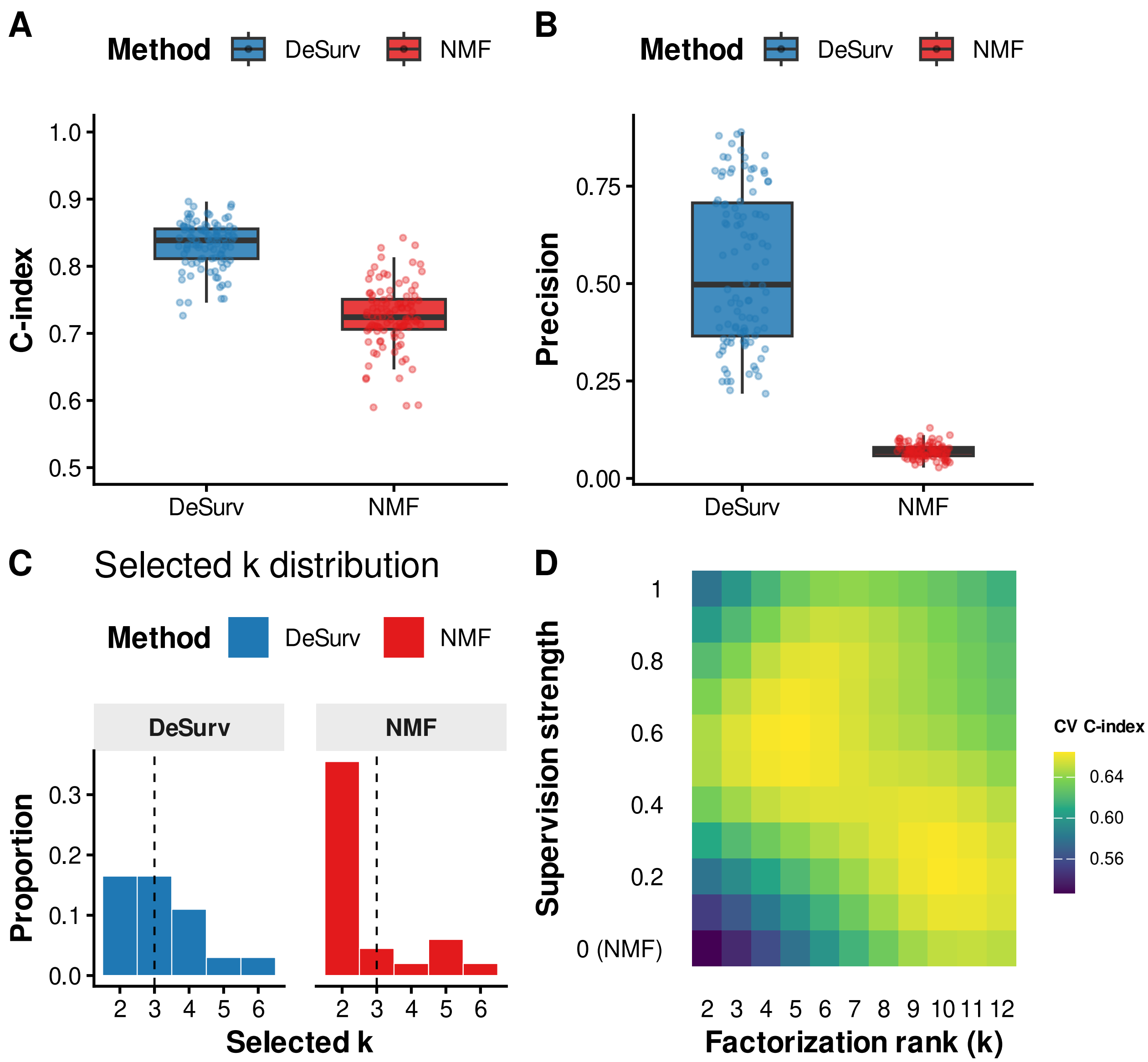

Fig. 2. Simulation studies and model selection. (A) Test-set concordance index across 100 simulation replicates ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$) comparing DeSurv to standard NMF ($\alpha = 0$); paired Wilcoxon $P < 0.001$. (B) Precision (fraction of a factor's top-weighted genes — defined in main text — overlapping a true prognostic program) across replicates. In (A) and (B), boxes show median and interquartile range (IQR); whiskers extend to $1.5\times$IQR; points beyond are individual replicates. The y-axis in (A) is truncated at 0.5 because C-index values below 0.5 indicate worse-than-random concordance. (C) Distribution of selected k values across the same 100 replicates for DeSurv versus standard NMF. (D) Gaussian-process–predicted mean cross-validated C-index from Bayesian optimization over factorization rank (k) and supervision strength (α) in PDAC training data (TCGA + CPTAC).

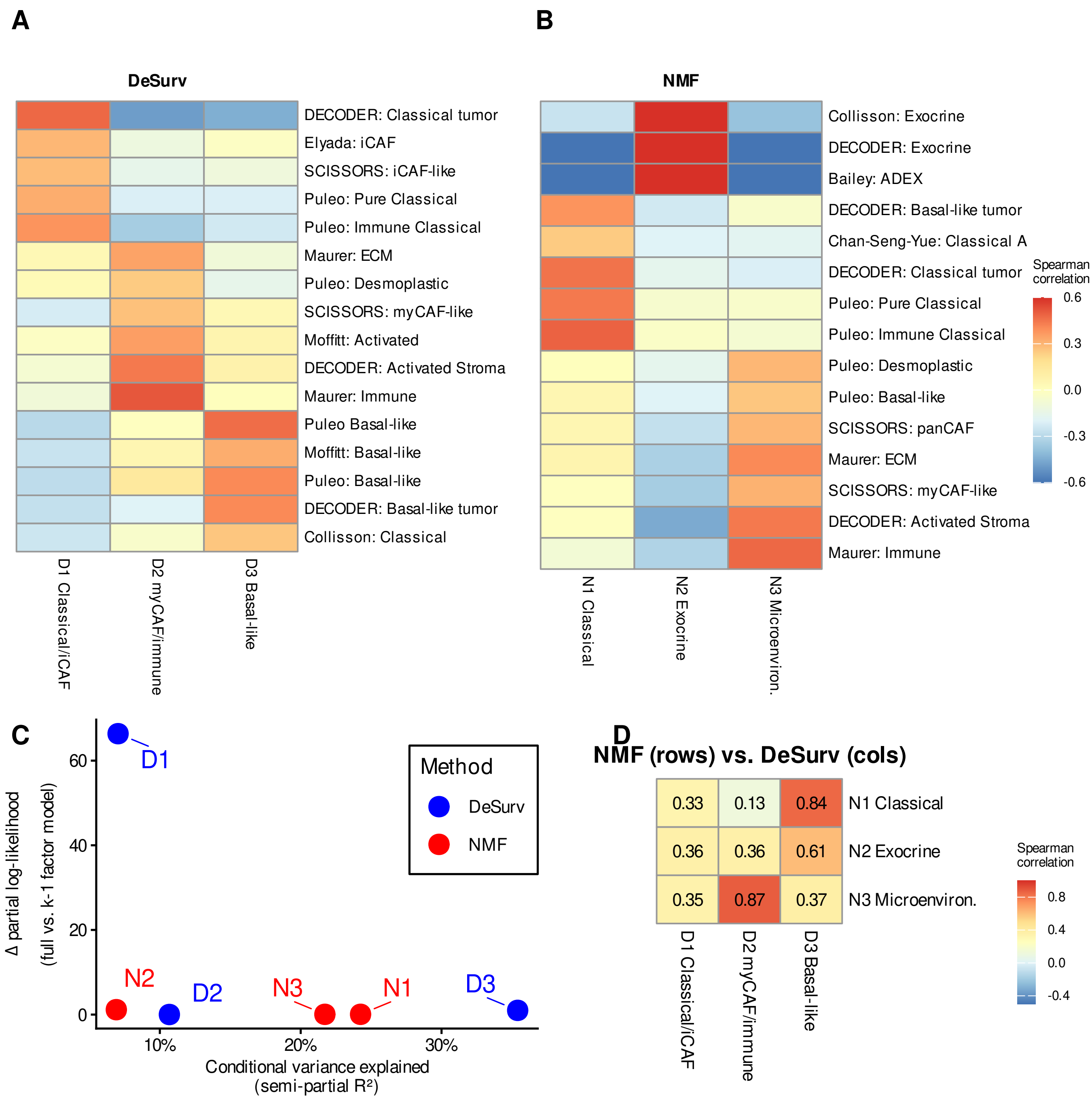

Fig. 3. Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (A–B) Spearman correlations between factor gene rankings and established PDAC gene programs for DeSurv (A) and standard NMF (B) at $k = 3$. Only $|r| > 0.2$ shown. DeSurv concentrates survival signal into D1 (Classical tumor + iCAF-associated stroma) while suppressing exocrine variation; standard NMF devotes one factor to exocrine signals. (C) Fraction of expression variance explained versus survival contribution (Δ Cox partial log-likelihood upon factor removal) for each factor. DeSurv concentrates survival signal into D1 while the highest-variance NMF factor (N2, exocrine-associated) contributes negligibly to survival. (D) Pairwise correspondence between NMF and DeSurv factors, measured by Spearman rank correlation of W-matrix gene loadings across all genes.

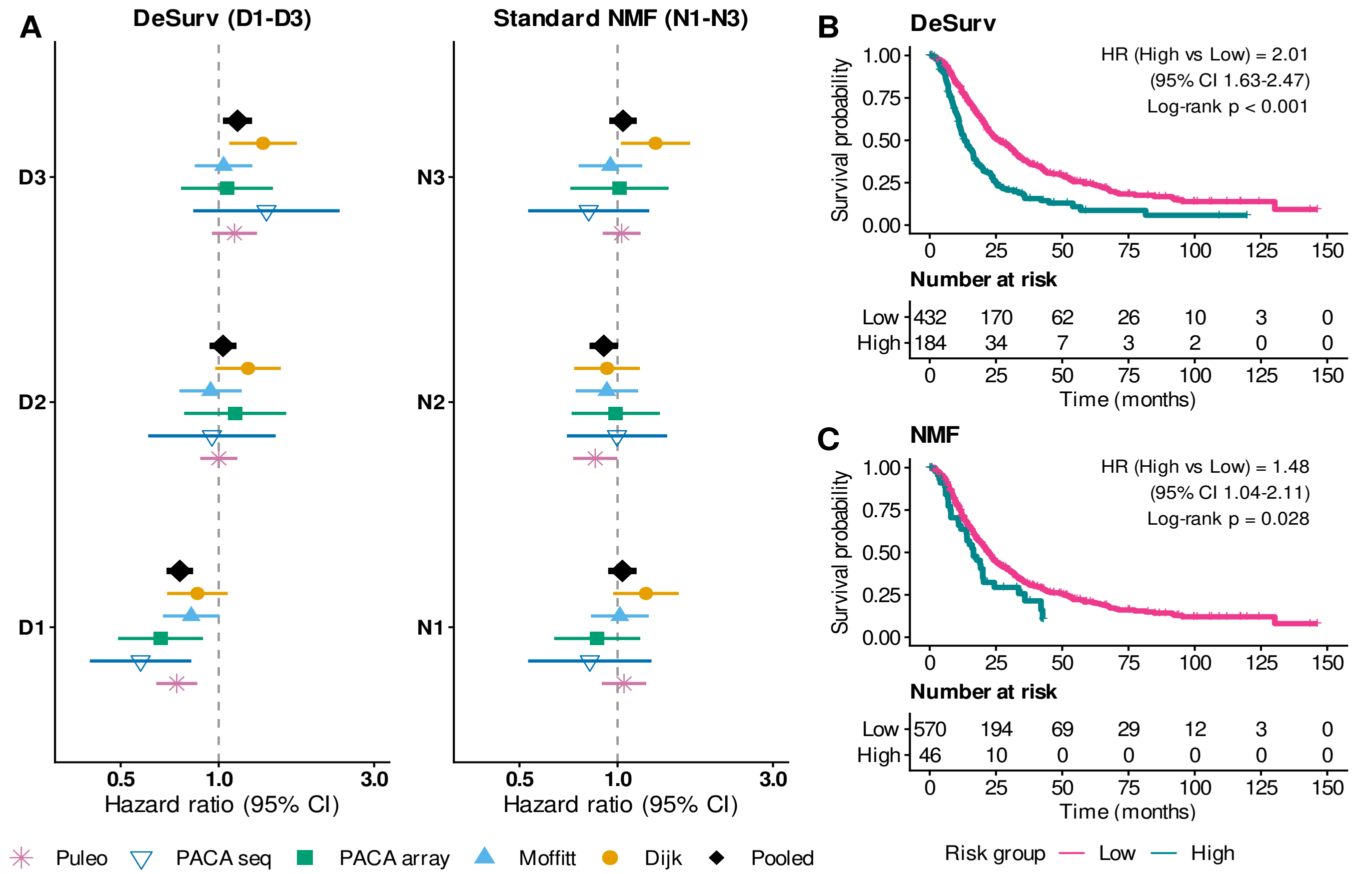

Fig. 4. Survival-aligned programs generalize across independent PDAC cohorts (pooled $n = 616$ patients across 5 cohorts: Dijk $n = 90$; Moffitt GEO array $n = 123$; PACA-AU array $n = 63$; PACA-AU RNA-seq $n = 52$; Puleo array $n = 288$). (A) Forest plot of univariate hazard ratios (95% CIs) from separate Cox models for each factor, showing each factor's marginal survival association in each validation cohort. DeSurv factor D1 shows a consistent protective association across all five cohorts (pooled HR = 0.76), while no standard NMF factor achieves a comparable pooled effect. Error bars are 95% confidence intervals; black diamonds denote inverse-variance-weighted pooled estimates. (B–C) Kaplan–Meier curves for pooled external validation samples stratified by the combined multivariate linear predictor ($\hat{\beta}_1 Z_1 + \hat{\beta}_2 Z_2 + \hat{\beta}_3 Z_3$, where coefficients are learned from training data), dichotomized at a cross-validated optimal cutpoint: DeSurv (B) and standard NMF (C).